

ME23312 ENGINEERING THERMODYNAMICS

Unit I – Basics, Zeroth and First Law · Presentation 3 of 10

Thermodynamic Equilibrium, Quasi-static and Reversible Processes

Mechanical, thermal and chemical equilibrium · the quasi-static process · reversible and irreversible processes · causes of irreversibility · free expansion

Dr. Anantha Raman L., M.E., Ph.D. · Associate Professor, Department of Mechanical Engineering, Rajalakshmi Institute of Technology, Chennai

In this presentation

1. What thermodynamic equilibrium means
 2. Mechanical, thermal and chemical equilibrium
 3. Why equilibrium matters
 4. Sudden versus slow compression
1. The quasi-static process
 2. Reversible processes
 3. Irreversible processes and their causes
 4. Free expansion

1

Thermodynamic equilibrium

What equilibrium means

- Properties do not change with time
- No tendency to change when the system is isolated from its surroundings
- Three conditions must be satisfied **simultaneously**:
mechanical, thermal and chemical equilibrium

The three conditions of equilibrium

Mechanical

No unbalanced forces within the system or with the surroundings; pressure uniform (allowing for gravity); no part being accelerated.

Thermal

Temperature uniform throughout and equal to that of the surroundings in contact – no heat flow.

Chemical

No chemical reaction; no diffusion or transfer of matter between parts; no phase change in progress.

Why equilibrium matters

- Classical thermodynamics deals **only with equilibrium states**
- A non-uniform system has no single value of pressure or temperature
- Properties can be defined only when the system is in equilibrium

KEY POINT No equilibrium $\Rightarrow$ no defined properties $\Rightarrow$ no point on the property diagram.

2

The quasi-static process

Infinitely slow – a succession of equilibrium states

Sudden versus slow compression

Sudden compression

- Pressure near the piston rises before the rest of the gas is affected
- Gas is **not in equilibrium** during the process
- Intermediate states cannot be plotted

Very slow compression

- Remove tiny weights one at a time
- Departure from equilibrium is infinitesimal at every instant
- System passes through a continuous succession of **equilibrium states**

The quasi-static (quasi-equilibrium) process

- An infinitely slow process in which the system is at every instant infinitesimally close to equilibrium
- Every intermediate state is a defined equilibrium state – the process can be **drawn as a continuous curve** on a property diagram
- Work can be evaluated as $\int p \, dV$ using the system pressure – the basis of all work expressions in this unit

KEY POINT Work = $\int p \, dV$ is valid only for a quasi-static process.

An idealisation that real processes approach

- Real processes approach the quasi-static ideal when the change is slow compared with the time the system needs to reach equilibrium
- For gases this **relaxation time** is very short
- Many real compressions and expansions can therefore be treated as quasi-static

3

Reversible and irreversible processes

Reversible process

- Can be reversed so that **both** system and surroundings are restored to their initial states – leaving **no trace** anywhere
- Passes through equilibrium states; forward and reverse heat and work transfers are equal and opposite
- Therefore **quasi-static and free of dissipative effects**
- The ideal standard: maximum work output, minimum work input; never fully achieved in practice

Irreversible process

- Cannot be reversed without leaving some change in the surroundings
- **All natural processes are irreversible**; the reversible process is the limiting ideal against which they are judged
- Internal irreversibility – within the system; external – at the boundary

Causes of irreversibility

Friction

Solid friction, fluid friction (viscosity), current through a resistance – work is converted to heat that cannot be fully recovered.

Unrestrained expansion

A gas expanding into a vacuum does no work; compressing it back needs work from the surroundings.

Heat transfer across a finite ΔT

Heat cannot flow back from the colder body to the hotter body without external work.

Mixing and others

Mixing of different substances or states, spontaneous chemical reactions, inelastic deformation, magnetic hysteresis.

Free expansion

- Rigid insulated vessel divided by a partition: gas on one side, vacuum on the other; the partition is removed
- No work – no resisting force at the moving boundary, no piston displaced
- No heat – the vessel is insulated $\Rightarrow$ internal energy unchanged; for an ideal gas the temperature is unchanged
- Irreversible: the gas never spontaneously collects itself back into one half

KEY POINT Free expansion is not quasi-static, so $\int p \, dV$ cannot be used – the work is simply zero.

Summary

- Equilibrium = mechanical + thermal + chemical, all at once
- Properties are defined only for equilibrium states
- Quasi-static: infinitely slow, succession of equilibrium states, can be drawn on a diagram
- Only for a quasi-static process is $W = \int p \, dV$
- Reversible = quasi-static + no dissipation; leaves no trace
- Irreversibility: friction, free expansion, finite- ΔT heat transfer, mixing
- Free expansion: $W = 0$, $Q = 0$, $\Delta U = 0$, but irreversible

Next: Zeroth law and temperature

- Thermal equilibrium and the Zeroth law
- Thermometers and thermometric properties
- The triple point and the Kelvin, Celsius, Fahrenheit and Rankine scales